Statistical and Computational Models

Lecture 14: Hierarchical models and MRP

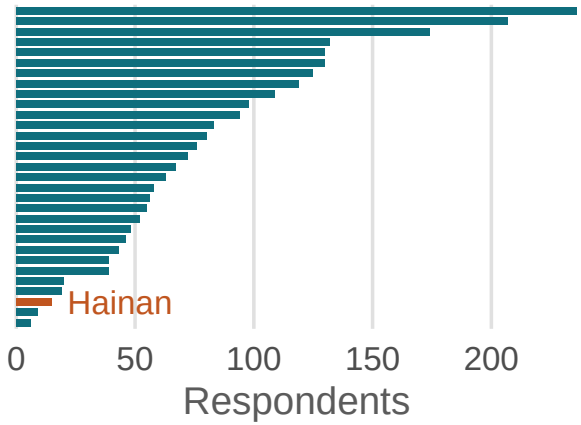
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Our simulated survey

Recall our survey of 2,500 adults across China's 31 provinces:

province	age	educ	hukou	trust
Yunnan	35-59	Secondary	Rural	1
Shanxi	18-34	College	Rural	0
Guangdong	35-59	Secondary	Urban	1
Jiangsu	18-34	College	Urban	1
Guangdong	35-59	College	Urban	0
Hainan	18-34	College	Urban	1



For each respondent we know whether they trust local government, and three traits (age, education, hukou) that the census counts for every adult.

At the province level, sample sizes run from 6 respondents (Tibet) to 236 (Guangdong).

Sub-national variation in trust

Previously we wanted to estimate trust in local government for Hainan. Now let's try to generalize this question: can we estimate trust in **every** province?

$Y_{ij} = 1$ if respondent i in province j trusts, and $\pi_{ij} = \mathbb{P}(Y_{ij} = 1)$. We want θ_j, province j's share who trust. First, let's just calculate the mean in each province:

Rank	Province	Respondents	Yes	Mean trust
1	Hainan	15	12	0.80
2	Ningxia	20	16	0.80
3	Hebei	130	103	0.79
...				
29	Hubei	98	34	0.35
30	Qinghai	9	3	0.33
31	Zhejiang	119	38	0.32

How seriously should we take this table as an answer to our question?

Roadmap

Partial pooling

Poststratification

MRP

Complete or no pooling

COMPLETE POOLING

One intercept for everyone:

$$\text{logit}\, \pi_{ij} = \beta_0$$

In R: `trust ~ 1`

The overall mean:
$0.58 = \text{logit}^{-1}(0.32)$.

Assumes the provinces are **all alike**.

NO POOLING (FIXED EFFECTS)

One intercept per province:

$$\text{logit}\, \pi_{ij} = \beta_{0j}$$

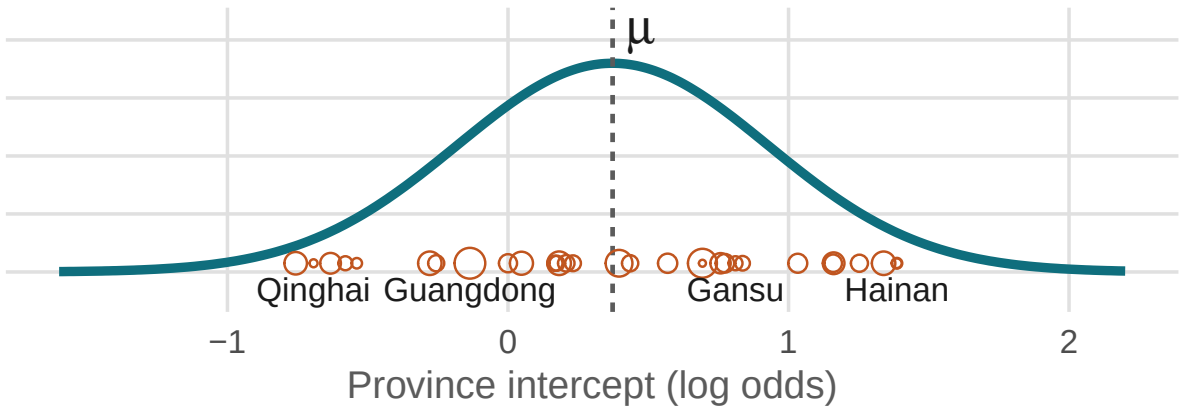
In R: `trust ~ 0 + province`

Hainan's own mean:
$0.80 = \text{logit}^{-1}(1.39)$.Qinghai's own mean:
$0.33 = \text{logit}^{-1}(-0.69)$.Guangdong's own mean:
$0.47 = \text{logit}^{-1}(-0.14)$.Gansu's own mean: $0.69 = \text{logit}^{-1}(0.81)$.

Assumes the provinces share **nothing**.

Fixed effects are very popular. But usually as nuisance parameters in a

Partial pooling



Consider the 31 fixed effects, one per province, in log odds ($\hat{\beta}_{0j}$) and scaled by sample size.

Now overlay **a Normal**, and assume $\beta_{0j} \sim N(\mu, \sigma_u^2)$.

Its centre μ is the **typical province** (not respondent), and its width σ_u how much the β_{0j} vary, if the model is right. Province j's shift is u_j: $\beta_{0j} = \mu + u_j$.

What do we get at $\sigma_u = 0$? Complete pooling. As $\sigma_u \to \infty$, no pooling.

Random effects

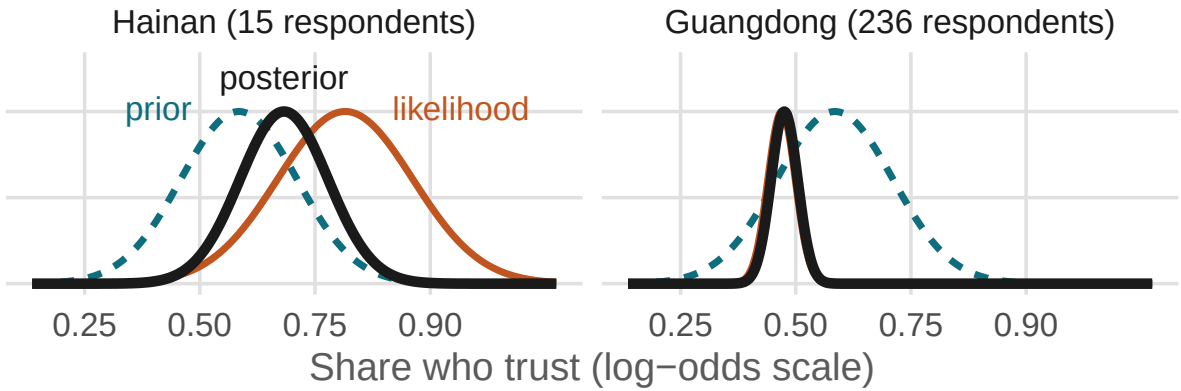
The shifts u_j are the **random effects**. Why "random"? The model treats each β_{0j} as an independent (random) draw from the **same Normal**.

Step 1: **estimate the Normal**, μ and σ_u, by maximum likelihood. $\hat{\sigma}_u = 0.55$ is smaller than the sd of the fixed effects, 0.66, because each fixed effect also carries sampling noise.

Step 2: **estimate each** β_{0j} by Bayes' rule, with the fitted Normal as its prior:

$$p(\beta_{0j} \mid \mathbf{Y}_j) \propto \underbrace{L(\beta_{0j};\, \mathbf{Y}_j)}_{\text{province } j\text{'s data}} \quad \underbrace{p(\beta_{0j})}_{\text{prior: } N(\hat{\mu},\, \hat{\sigma}_u^2)}$$

Estimating each province's π



Step 1 gives the **prior**, the same fitted Normal for every province. The mean for every province is most likely to be near $\hat{\mu}$.

Step 2 gives each province's **likelihood**, centred on its own mean.

The **posterior**, prior $\times$ likelihood, will always fall between the province-specific $\hat{\beta}_{0j}$ and $\hat{\mu}$. This is **partial pooling**, or **shrinkage**: a little bias toward $\hat{\mu}$, for much less variance.

It is the familiar prior-times-likelihood update, now estimated for every province at once, with the prior estimated from the data: **empirical Bayes**.

Fitting a hierarchical logit in R

Using `lme4::glmer()`:

```
fit_pp <- lme4::glmer(trust ~ 1 + (1 | province), data = respondents,
                      family = binomial)
round(coef(summary(fit_pp))[, 1:2, drop = FALSE], 2)
```

```
            Estimate Std. Error
(Intercept)     0.37       0.11
```

```
print(lme4::VarCorr(fit_pp), digits = 2)
```

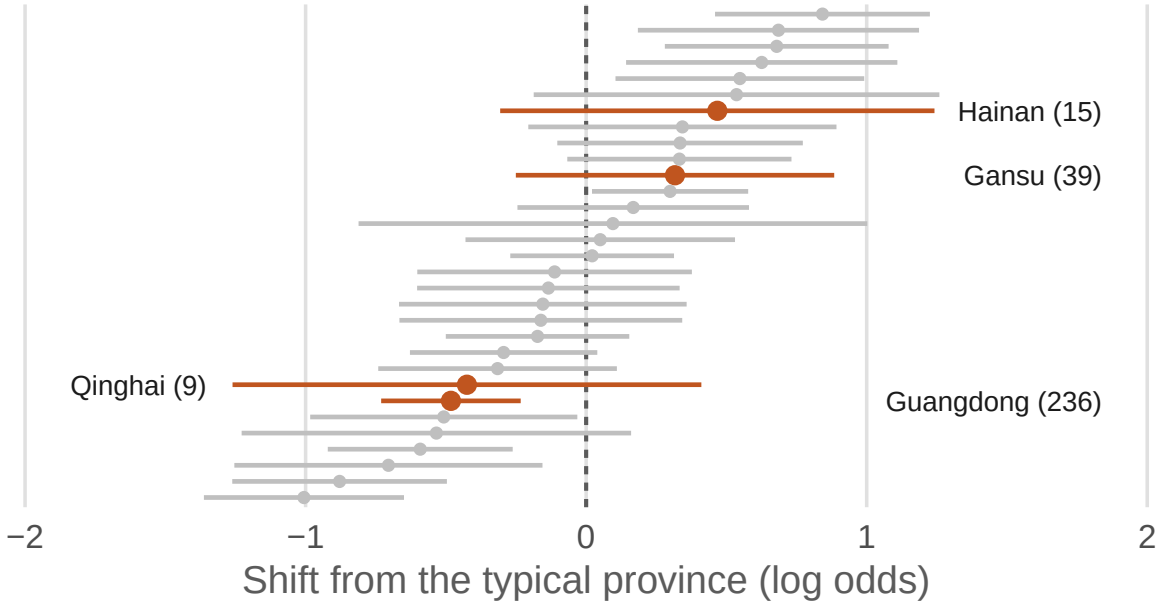
```
 Groups   Name        Std.Dev.
 province (Intercept) 0.55
```

Note: `(1 | province)` gives one intercept per province, from one Normal.

The one row, `(Intercept)`, is $\hat{\mu}$: $\text{logit}^{-1}(0.37) = 0.59$ is the **typical province** (not 0.58: provinces count about equally). The last line is $\hat{\sigma}_u = 0.55$, in log odds.

`lme4::ranef()` predicts the 31 $\hat{u}_j$ (predicted: they are not parameters).

The predicted random effects



`lme4::ranef(fit_pp)` gives each province's shift from the typical province, $\hat{u}_j$. Above 0, more trusting than typical; below, less.

Each interval is $\hat{u}_j \pm 1.96$ sd, the sd of province j's posterior (`condsd` in `as.data.frame(ranef(fit_pp))`).

They treat $\hat{\mu}$ and $\hat{\sigma}_u$ as known, so they are a little too narrow.

Roadmap

Partial pooling

Poststratification

MRP

Selection bias

Most surveys, no matter how they are constructed, end up suffering from **survey selection bias**.

In our case, compared against China's census, urban, educated and younger adults respond more often:

Trait	Respondents	All adults
Urban hukou	68%	48%
College	37%	19%
Aged 60+	14%	23%

If selection is correlated with our outcome, we will recover inaccurate estimates of the (descriptive) parameters of interest.

In our case, urban, educated and younger adults tend to trust less, so our survey underrepresents trust both nationally and within the provinces.

Poststratification

Sort all our respondents, across China, into **cells** of people with the same traits (e.g. hukou). The census gives each cell's **population target**, its share of China's adults:

Hukou	Share who trust	Share of respondents	Population target	Weight
Urban	0.53	68%	48%	0.71
Rural	0.69	32%	52%	1.62

Each respondent gets a **poststratification weight**: target over share of respondents. With these weights, the sample share moves:

$$\begin{aligned} \text{Unweighted:} \quad & 0.68 \times 0.53 + 0.32 \times 0.69 = 0.58 \\ \text{Weighted:} \quad & (0.68 \times 0.71) \times 0.53 + (0.32 \times 1.62) \times 0.69 \\ & = 0.48 \times 0.53 + 0.52 \times 0.69 = 0.61 \end{aligned}$$

In general, with s_c a cell's population target and $\bar{y}_c$ its sample share: $\hat{\theta} = \sum_c s_c \bar{y}_c$.

Province by province

To estimate each province's θ_j, poststratify **within** it:

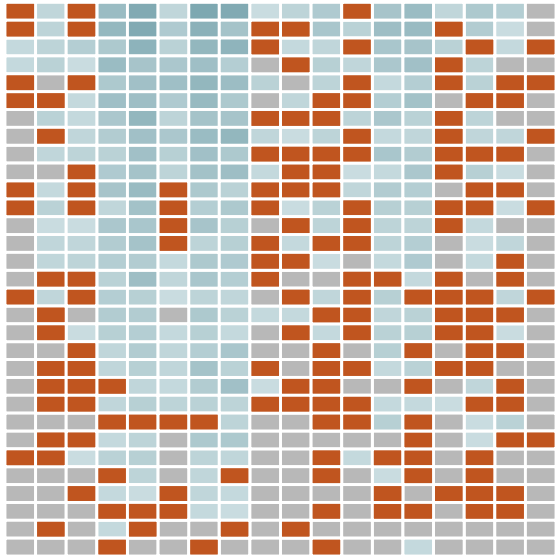
	Urban			Rural		
	Primary	Secondary	College	Primary	Secondary	College
Guangdong	8	65	85	20	39	19
Gansu	7	13	9	5	4	1
Hainan	0	4	5	1	3	2
Qinghai	2	4	2	1	0	0

In Qinghai, there is only a single rural respondent. To bring the Qinghai sample back to the Qinghai target (54% rural) we have to give that respondent a weight of 4.9. The estimate of trust falls from 0.33 to 0.17!

Split by a second dimension – education – and the cells thin out dramatically. Hainan and Qinghai now have cells with **no respondent**, so their poststratified shares cannot be computed.

Sparse cells

Now cross all three traits, age, education and hukou: 18 cells in each province, 558 across all 31. Many are nearly empty:



137 (25%) of all cells have no respondent at all – **sparsity**.

In 169 more, there is no variation in Y – **separation**.

This is the small-province problem again, at a finer level. The fix is also the same: **partial pooling**, across cells and provinces.

Roadmap

Partial pooling

Poststratification

MRP

Features at two levels

We want to borrow both across provinces and across cells. To do that well, we need information about the cells, and also about the provinces: **features at both levels**. So each respondent now also carries their province's `lgdp`:

province	age	educ	hukou	lgdp	trust
Hainan	18-34	College	Urban	−0.20	1
Hainan	18-34	College	Rural	−0.20	1
Guangdong	35-59	Secondary	Urban	0.18	0
Guangdong	18-34	Primary or less	Rural	0.18	1
Qinghai	35-59	College	Urban	−0.27	1
Gansu	35-59	Primary or less	Urban	−0.62	1

`lgdp` is **engineered**: log GDP per capita relative to the national 89,000 yuan, so 0 is average and Hainan's −0.20 is about 18% below.

`age`, `educ` and `hukou` vary **within** a province; `lgdp` only **between** provinces. It is a province-level feature, entered for every respondent.

A multilevel model

Model $\pi_{ij} = \mathbb{P}(Y_{ij} = 1)$, respondent i's chance of trusting, with **features from both levels**: their traits and their province's GDP.

$$\text{logit}\, \pi_{ij} = \beta_{0j} + \underbrace{\beta_1 x_{ij1} + \cdots + \beta_5 x_{ij5}}_{\text{age, educ, hukou of respondent } i}$$

$$\beta_{0j} \sim N\Big(\underbrace{\beta_0 + \beta_{\text{lgdp}}\, \text{lgdp}_j}_{\text{the centre } \mu\text{, now a line in GDP}}, \sigma_u^2\Big)$$

Read, respondent level: each trait shifts the log odds as in any logit, by the same amount in every province.

Province level: each province's β_{0j} is still drawn from a Normal, but its **centre** is set by β_0, the typical province at average GDP, and by the province's own lgdp_j.

So $\beta_{0j} = \beta_0 + \beta_{\text{lgdp}}\, \text{lgdp}_j + u_j$, and u_j is what GDP leaves unexplained.

A multilevel model in R

Fitting it with `lme4::glmer()`: the traits and `lgdp` enter like any other feature.

```
fit_hier <- lme4::glmer(
  trust ~ age + educ + hukou + lgdp + (1 | province),
  data = respondents, family = binomial)
round(lme4::fixef(fit_hier), 2)
```

```
  (Intercept)      age35-59        age60+ educSecondary   educCollege
         1.04          0.27          0.36         -0.54         -0.78
  hukouUrban          lgdp
        -0.52         -1.02
```

The rows are the **coefficients** $\hat{\beta}$, which linearly map $\boldsymbol{x}_{ij}$ to the log odds, with $\boldsymbol{u}_j$ held fixed. `lme4` calls these the fixed effects – for our own clarity we might call them **common or non-varying effects** instead.

With traits and GDP, $\hat{\sigma}_u$ falls from 0.55 to 0.37: the features explain much of how provinces differ, so a small province leans more on what is **typical for provinces like it**.

Multilevel regression and poststratification

Now combine the model with poststratification. This is **MRP**:

1. Fit a hierarchical logit of trust on respondent traits, province-level features, and province intercepts.
2. Predict every cell c (one combination of traits in one province), empty cells included:

$$\hat{\pi}_c = \text{logit}^{-1}\left(\boldsymbol{x}_c^\top \hat{\boldsymbol{\beta}} + \hat{u}_j\right)$$

3. Poststratify each province: $\hat{\theta}_j^{\text{MRP}} = \sum_{c \in j} s_c \, \hat{\pi}_c$.

More generally, MRP works for any units nested in groups: voters in districts, students in schools. And the model can be more flexible: we could estimate trust in each prefecture, nested within provinces, with respondent traits whose effects vary by province (**varying slopes**).

MRP in R

`avg_predictions()` from `{marginaleffects}` does steps 2 and 3:

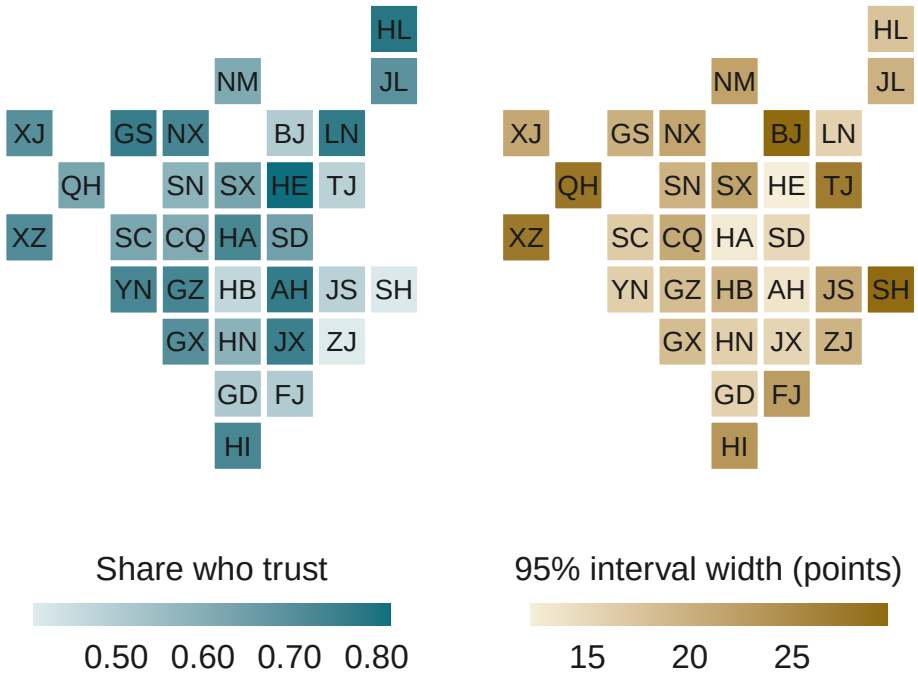
```
marginaleffects::avg_predictions(
  fit_hier, by = "province", wts = "share",
  newdata = subset(census, province %in%
                     c("Hainan", "Qinghai", "Guangdong", "Gansu")))
```

```
  province Estimate Std. Error 2.5 % 97.5 %
 Gansu        0.765     0.0268 0.713  0.818
 Guangdong    0.512     0.0251 0.463  0.561
 Hainan       0.732     0.0173 0.698  0.766
 Qinghai      0.629     0.0220 0.586  0.672
```

Note: we keep the province intercepts in. `re.form = NA` would set every $\hat{u}_j$ to 0, leaving each province only what its GDP and its census mix predict.

Warning: these standard errors treat every $\hat{u}_j$ as known, so they are too **narrow**. Here 2 of the 4 intervals miss the true share. A fully Bayesian fit carries that uncertainty through: `rstanarm::stan_glmer()` takes the same formula.

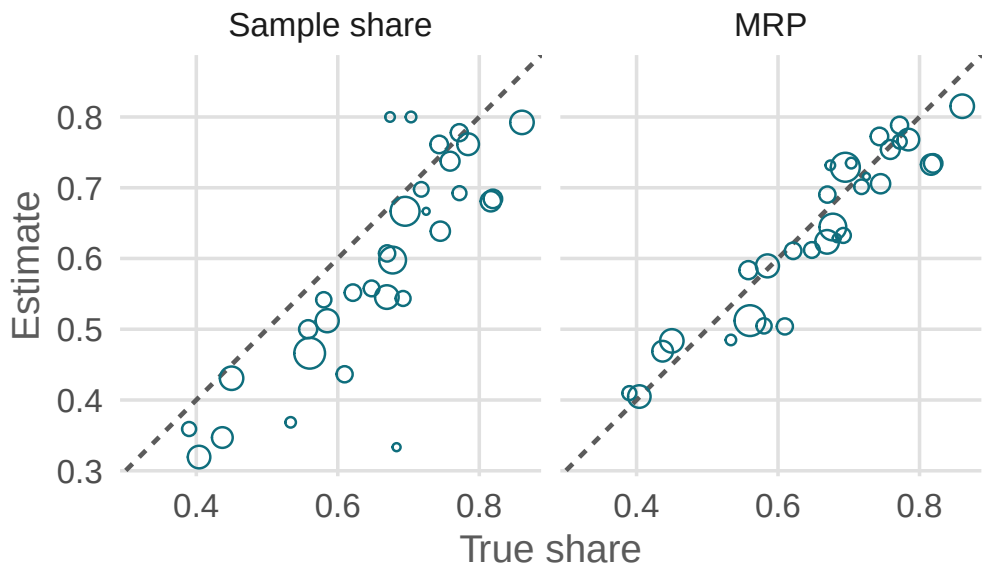
Mapping trust



MRP's estimate of trust in every province, adjusted for who answered.

Its uncertainty: each 95% interval's width, simulated to include each province's $\hat{u}_j$. Widest for the smallest provinces, and for the richest, at the edge of GDP's range: Beijing and Shanghai.

How did we do?



Each province's estimate against its true share, known only because we simulated.

Most sample shares sit too **low**: selection bias, plus noise in the small provinces.

MRP sits close to the truth. Hainan: 0.80 by its sample share, 0.73 by MRP, truly 0.67. Qinghai: 0.33, 0.63, truly 0.68.

Checking the model

In a real survey we never see θ_j. We can still check the model the usual way, by **cross-validation**: test MSE of each respondent's predicted Pr(trust), against their answer.

Model	New respondents	New province
Complete pooling	0.244	0.245
Partial pooling	0.230	0.245
MRP model	0.222	0.227

The MRP model predicts best, for new respondents and for a province it has never seen. The gaps look small because a yes-or-no answer is mostly noise.

But held-out respondents were selected like the rest, so cross-validation **cannot see selection bias**. Correcting it rests on assumptions.

What MRP assumes

MRP assumes:

1. Each trait's effect is **common** to every province (here, by construction). If hukou mattered more in some provinces than others, we would need varying slopes.
2. The census counts each province's population in every cell: their **joint** mix of traits.
3. Within a cell, those who answered are like those who did not (here, by construction).
4. No province is too unusual: a truly unusual province is pulled toward the model too.

What is the estimand?

A LEVEL

θ_j: trust in each province, or in any subgroup, such as rural adults over 60.

The intercepts are **the target**. Random effects borrow across provinces to estimate them.

A RELATIONSHIP

β_{urban}: how much less urban-hukou adults trust.

The intercepts are **a nuisance**. Fixed effects are often a safe choice, as province-level factors may be confounders.

The reason: random effects assume each u_j is **unrelated** to the traits. If more urban provinces trusted less for some other reason, $\hat{\beta}_{\text{urban}}$ would absorb it. Fixed effects compare respondents **within** a province, so they cannot.

Here the assumption holds by construction: $\hat{\beta}_{\text{urban}} = -0.52$ either way. For a middle way between the two, see the Mundlak device and correlated random effects models (Mundlak 1978; Bell & Jones 2015).

References

- Bell, A. & Jones, K. (2015) “Explaining Fixed Effects: Random Effects Modeling of Time-Series Cross-Sectional and Panel Data.” *Political Science Research and Methods* 3(1): 133–153.
- Caughey, D. & Warshaw, C. (2019) “Public Opinion in Subnational Politics.” *Journal of Politics* 81(1): 352–363.
- Johnson, A. A., Ott, M. Q. & Dogucu, M. (2022) *Bayes Rules! An Introduction to Applied Bayesian Modeling*. CRC Press – **ch. 18**. bayesrulesbook.com.
- Mundlak, Y. (1978) “On the Pooling of Time Series and Cross Section Data.” *Econometrica* 46(1): 69–85.